

Supplementary Information

DrugDis: Disentangling general and context-specific effects in drug-response prediction

This Supplementary Information contains three Supplementary Notes on the response decomposition, zero-amplitude predictions and cross-assay reproducibility; 22 Supplementary Tables with detailed benchmark results; and one Supplementary Data item describing the data sources.

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Supplementary Notes

Note 1: the decomposition on an unbalanced observed support

The projection H is fixed by the observed support alone, and the support of a pharmacogenomic dataset is incomplete and unbalanced. Three properties of the decomposition follow.

The drug and sample marginals share part of the additive variance. On a complete balanced grid the two sets of indicator columns are orthogonal, and the additive variance splits uniquely into a drug part and a sample part. On an unbalanced support they are correlated, so the additive variance divides into a part unique to each marginal and a shared part. On the benchmark dataset the drug marginal alone accounts for 72.65% of total response variance and the sample marginal alone for 4.17%, and 0.97% is shared. The unique parts are 71.68% and 3.20%, and the three sum to the 75.85% the additive component carries. Across the ten test sets the shared part ranges from 0.46% to 1.40% (Supplementary Table 2). The main text quotes the unique shares.

The interaction component is indexed by the support that defines it. Two evaluation sets with different supports define different projections, so each defines its own interaction component. The dimension of the additive subspace is $\text{rank}(H) = n_{\text{drug}} + n_{\text{sample}} - c$ with c the number of connected components of the bipartite support graph, and $c = 1$ for the benchmark dataset and for every test set. The fraction of degrees of freedom the projection removes varies by an order of magnitude between evaluation sets: 1.76% on the benchmark dataset, 9.52% to 11.45% on the held-out cell-line test sets, and 1.92% to 1.98% on the held-out compound test sets. The error density in Methods is therefore normalized by the rank of each subspace, and component values are compared within a distribution-shift regime. A compound or sample observed once has its response as its own marginal, so the projection assigns that response entirely to the additive component; sparsely observed compounds and samples contribute to the additive component by construction.

The marginals reflect the screening design as well as the biology. The observed support covers 3,141,680 of the $54,180 \times 986$ possible drug-sample pairs, 5.88% of the grid, and the screened pairs were chosen by the resources that ran them. The drug marginal is the mean response of a compound over the samples on which it was screened, and the sample marginal is the mean response of a sample over the compounds applied to it. The additive component is therefore defined by the observed design as well as by pharmacology, and the interaction component is the deviation from the best additive fit to that design. Every component value in this work is stated on a named support: the benchmark dataset, a test set or a single response resource (Supplementary Table 3).

Note 2: the degenerate case of the interaction-component correlation

The interaction-component correlation is $\text{corr}(My, M\hat{y})$. When the prediction lies in the additive subspace, $M\hat{y} = 0$, both the numerator and the denominator of the Pearson formula vanish, and the quantity is undefined. This is the situation of every predictor that carries no drug-sample interaction information.

Which predictors fall in it. A constant predictor does, and so does any predictor assembled from drug-level and sample-level terms, including one built from training-set marginals. A structurally additive decoder does as well, for the same reason. Across the 28 runs of the additive ridge decoder in the decoder comparison, the standard deviation of the emitted interaction lies between 1.6×10^{-9} and 1.2×10^{-8} , against an observed interaction standard deviation of order 7×10^{-2} , and the emitted amplitude A_{int} is zero to six decimal places in every one (Supplementary Table 11). The Pearson formula returns -0.019 to $+0.017$ for those runs, the correlation of floating-point noise, so the main text compares these decoders by emitted amplitude (Fig. 3g). Methods define the threshold, $A_{\text{int}} \leq 10^{-8}$, below which a prediction is classed as emitting no interaction and its interaction-component correlation is reported as undefined (Supplementary Table 5).

Shuffled predictions set the null scale. A predictor that emits interaction but carries none of its structure gives a measured zero on a defined quantity. Shuffled predictions emit 1.3480 to 1.9275 times

the amplitude the observed interaction contains and score -0.001 under held-out cell lines, $+0.000$ under held-out compounds and $+0.000$ in organoids, which calibrates the scale of the interaction-component correlation.

Note 3: interpretation of the cross-assay reproducibility reference

Cross-assay reproducibility q_c is the correlation between GDSC1 and GDSC2 measurements of the same drug-sample pair, computed component by component with one projection built on the matched support and applied identically to both arms. On the 32,724 matched pairs that meet the benchmark eligibility rules it is 0.650 for total response, 0.754 for the additive component and 0.498 for the interaction component. The empirical reproducibility reference is $\sqrt{q_c}$, that is 0.807, 0.869 and 0.706, and recovery relative to it is $\eta_c = r_c / \sqrt{q_c}$ (Supplementary Table 6).

The square root follows from a classical measurement model. If two measurements obey $y_A = z + \epsilon_A$ and $y_B = z + \epsilon_B$, where z is a common latent quantity, the errors have equal variance, are mutually independent and are independent of z , then $q_c = \text{Var}(z) / \text{Var}(y_A)$ and a perfect predictor of z has correlation $\sqrt{q_c}$ with one measured arm. GDSC1 and GDSC2 differ in assay generation, so their errors need not be exchangeable or independent, and $\sqrt{q_c}$ serves as an empirical comparison reference.

Analysis supports. The analyses use the following subsets of matched observations, all on DROMA Sensitivity without rescaling.

Stage	Pairs and transformation	Count	Use
Benchmark response panel	re- Canonical SMILES and DROMA sample identifier; mean of the retained measurements	3,141,680	Training and evaluation target for the benchmark.
GDSC1-GDSC2 match	Exactly one GDSC1 and one GDSC2 measurement for the same canonical pair, giving y_A, y_B	43,622; 89 drugs, 799 samples	Starting set for cross-assay reproducibility and independent model-selection analyses.
Benchmark eligibility filter	Matched pairs that meet the benchmark eligibility rules	32,724; 89 drugs, 600 samples	Full-support q_c reference.
Exact recovery support	Matched test pairs whose benchmark target equals y_A within 10^{-6}	Held-out cell lines 2,876–3,051; held-out compounds 1,970–3,873 per seed	Support for η_c ; the target-equality filter excludes pairs whose benchmark target comes from another resource.
Independent model-selection support	Matched pairs in the test set of each split and seed, shared by M0 and M2	Held-out cell lines 4,927–5,411 (mean 5,138); held-out compounds 2,723–7,164 (mean 5,027)	GDSC1 selection metrics and GDSC2 independent endpoint.

The reference has three properties.

It measures reproducibility across assay generations. The two arms are GDSC1 and GDSC2, two assay generations run by one consortium, so the estimate combines measurement noise with the assay change.

It summarizes a magnitude-dependent reproducibility. On the matched pairs, q_{int} rises from 0.08 in the smallest quartile of interaction magnitude to 0.62 in the largest, and one scalar per component summarizes that range. Recovery ratios can therefore exceed one: under held-out cell lines the additive η averages 1.00 across five seeds, with a range of 0.97 to 1.03 (Supplementary Table 14). In that regime the benchmark target is the GDSC1 measurement and compounds are not held out, so compound-level structure specific to GDSC1 is available to the predictor.

Its uncertainty is dominated by the compound axis. Observations sharing a compound or a sample are correlated, so intervals come from cluster bootstraps. For total response the sample bootstrap returns

0.640 to 0.660 across 600 clusters, while the compound bootstrap returns 0.529 to 0.755 across 89. The two-way scheme, which resamples both axes with draw multiplicity retained, gives 0.528 to 0.750. Intervals on η_c resample both axes once per replicate and recompute the model correlation and the reference together, so the numerator and the denominator carry their uncertainty jointly.

On the exact recovery support the scored target equals the single arm-A measurement, so the reference applies without the Spearman–Brown rescaling $q_n = nq / (1 + (n - 1)q)$ that an average of n measurements would require (Spearman, *Br. J. Psychol.* **3**, 271–295, 1910; Brown, *Br. J. Psychol.* **3**, 296–322, 1910).

Supplementary Tables

Supplementary Table 1: Benchmark dataset composition. The transcriptomic input matrix is CCLE-derived, restricted to 15,961 genes and holds one profile per sample. A sample enters the benchmark dataset only if a CCLE baseline profile exists for it. Response rows are counted per resource after the overlap rule of the processed response table and before rows sharing a canonical drug-sample pair were averaged (Methods).

Item	Value
Transcriptomic input	CCLE, 15,961 genes
Profiled samples in the input matrix	1,406
Drug-sample pairs in the benchmark dataset	3,141,680
Cell-line samples	986
Compounds	54,180
NCI60 (response rows)	2,175,581
PRISM (response rows)	504,493
CTRP2 (response rows)	269,725
GDSC1 (response rows)	125,914
GDSC2 (response rows)	44,797
CTRP1 (response rows)	39,784
CCLE (response rows)	10,730
GRAY (response rows)	1,888
gCSI (response rows)	1,646
FIMM (response rows)	890
UHNBreast (response rows)	54
All response resources	3,175,502

Supplementary Table 2: Orthogonal response decomposition on the benchmark dataset and on every test set. $\text{rank}(H)$ is the dimension of the additive subspace and d_H/N the fraction of degrees of freedom it occupies, in per cent. Additive and Interaction are the percentages of total response variance carried by the two components. Drug and Sample are the percentages uniquely attributable to each marginal and Shared the percentage carried by both (Supplementary Note 1). $|\text{corr}|$ is the absolute correlation between the two components, in units of 10^{-11} .

Evaluation set	Seed	Pairs	Compounds	Samples	$\text{rank}(H)$	d_H/N	Additive	Interaction	Drug	Sample	Shared	$ \text{corr} $
Benchmark dataset	—	3,141,680	54,180	986	55,165	1.76	75.85	24.15	71.68	3.20	0.97	2.72
Held-out cell lines	3407	557,484	54,176	148	54,323	9.74	78.36	21.64	74.62	2.34	1.40	1.39
	3408	474,279	54,176	148	54,323	11.45	76.22	23.78	71.31	4.46	0.46	1.68
	3409	486,991	54,172	148	54,319	11.15	77.16	22.84	73.81	2.46	0.90	3.20
	3410	570,863	54,175	148	54,322	9.52	76.76	23.24	72.24	3.86	0.67	2.58
	3411	530,351	54,175	148	54,322	10.24	76.93	23.07	73.40	2.78	0.75	2.68
Held-out compounds	3407	468,871	8,127	986	9,112	1.94	74.78	25.22	70.73	3.19	0.86	2.09
	3408	466,230	8,127	984	9,110	1.95	75.83	24.17	71.82	2.98	1.03	2.04
	3409	459,058	8,127	984	9,110	1.98	75.93	24.07	71.56	3.37	1.00	2.62
	3410	475,001	8,127	986	9,112	1.92	75.82	24.18	71.64	3.31	0.87	1.01
	3411	474,086	8,127	984	9,110	1.92	76.17	23.83	72.45	3.19	0.53	0.57

Supplementary Table 3: Orthogonal response decomposition within each response resource. Each resource is decomposed on its own eligible screen: its DROMA measurements that meet the inclusion criteria of the benchmark dataset, with rows sharing a canonical drug-sample pair averaged within the resource and no value from another resource. The benchmark dataset row repeats Supplementary Table 2. Without NCI60 is the benchmark dataset rebuilt from the other ten resources by the rules that built it. Resources are ordered by the number of pairs in their screen. Columns as in Supplementary Table 2, in per cent. Shared is the inclusion-exclusion overlap of the two marginal subspaces, which are correlated on an unbalanced support, and can be slightly negative (Supplementary Note 1). ^bPairs and additive share on the rows that the benchmark dataset attributes to the resource, which are fewer where the same compound name and sample were also measured by a resource earlier in the order of the overlap rule (Methods). The absolute correlation between the two components is below 10^{-9} in every row.

Response support	Pairs	Compounds	Samples	d_H/N	Additive	Interaction	Drug	Sample	Shared	Pairs ^b	Additive ^b
Benchmark dataset	3,141,680	54,180	986	1.76	75.85	24.15	71.68	3.20	0.97	—	—
Without NCI60	993,482	1,926	984	0.29	66.54	33.46	60.59	4.13	1.83	—	—
NCI60	2,170,974	52,908	68	2.44	82.54	17.46	79.48	2.69	0.37	2,157,687	82.59
PRISM	612,596	1,375	473	0.30	81.27	18.73	79.37	1.90	-0.01	503,644	81.61
CTRP2	292,817	462	822	0.44	69.74	30.26	60.06	8.55	1.13	269,725	69.94
GDSC1	167,886	293	701	0.59	59.04	40.96	53.17	5.93	-0.06	125,914	58.17
GDSC2	88,507	174	605	0.88	59.03	40.97	52.37	6.66	0.00	44,797	56.01
CTRP1	40,085	307	234	1.35	73.17	26.83	57.82	14.04	1.31	39,784	73.32
gCSI	11,495	41	523	4.90	71.51	28.49	65.13	7.31	-0.93	1,646	83.50
CCLC	10,730	24	472	4.61	79.27	20.73	73.99	5.02	0.27	10,730	79.27
GRAY	3,087	95	50	4.66	72.75	27.25	65.04	7.84	-0.13	1,888	74.76
FIMM	2,345	52	46	4.14	56.59	43.41	46.73	9.80	0.06	890	58.13
UHNBreast	166	6	39	26.51	48.04	51.96	24.45	21.55	2.04	54	87.98

Supplementary Table 4: Behaviour of existing evaluation views and DrugDis under prediction perturbations. Entries are mean changes from the original prediction over M0 and M2 at five seeds. Each metric is computed on identical predictions and pair support within a run. Drug-only and cell-only offsets have a standard deviation equal to half that of the observed response. The two interaction perturbations retain the projected additive prediction and multiply only the projected interaction prediction.

Regime	Metric	Drug-only offset	Cell-only offset	Interaction $\times 0.5$	Interaction $\times 2$
Held-out cell lines	Global	-0.1185	-0.1216	-0.0015	-0.0209
	Fixed drug	0.0000	-0.1203	0.0111	-0.0471
	Fixed cell	-0.1056	0.0000	-0.0014	-0.0324
	DrEval normalized	-0.1544	-0.1663	-0.0485	0.0230
	DrugDis interaction correlation	-0.0000	0.0000	0.0000	0.0000
	DrugDis interaction amplitude	0.0000	-0.0000	-0.1807	0.3614
	DrugDis interaction R^2	-0.0000	0.0000	-0.0138	-0.1702
Held-out compounds	Global	-0.1230	-0.1207	-0.0030	-0.0216
	Fixed drug	0.0000	-0.2051	0.0021	-0.0950
	Fixed cell	-0.1111	-0.0000	0.0008	-0.0291
	DrEval normalized	-0.1254	-0.1232	-0.0035	-0.0213
	DrugDis interaction correlation	0.0000	0.0000	0.0000	0.0000
	DrugDis interaction amplitude	0.0000	-0.0000	-0.1833	0.3667
	DrugDis interaction R^2	0.0000	0.0000	-0.0088	-0.1842

Supplementary Table 5: Null-predictor calibration. Mean and standard deviation across five seeds for four predictors under each evaluation regime. Total, Additive and Interaction are the three component-wise correlations. Entries read *undefined* where the predicted vector carries no variance in the corresponding subspace: for the constant predictor in all three, and for the constant and marginal predictors in the interaction subspace, which is the case Supplementary Note 2 treats. $\text{sd}(M\hat{y})$ is the emitted interaction amplitude in absolute units and $\text{sd}(My)$ the amplitude the observed interaction carries. The organoid rows are the primary pooled set (UMPDO1–3).

Regime	Predictor	n	Total	Additive	Interaction	$\text{sd}(M\hat{y})$	$\text{sd}(My)$
Held-out cell lines	Constant	5	undefined	undefined	undefined	1.88×10^{-17}	0.0711
	Training marginals	5	0.849 (0.010)	0.967 (0.007)	undefined	1.29×10^{-18}	0.0711
	Shuffled predictions	5	-0.001 (0.001)	-0.001 (0.005)	-0.001 (0.002)	0.1203	0.0711
	Trained predictor (M0)	5	0.858 (0.009)	0.963 (0.007)	0.315 (0.040)	0.0247	0.0711
Held-out compounds	Constant	5	undefined	undefined	undefined	1.25×10^{-16}	0.0732
	Training marginals	5	0.196 (0.006)	0.226 (0.008)	undefined	1.22×10^{-10}	0.0732
	Shuffled predictions	5	0.000 (0.002)	-0.002 (0.011)	0.000 (0.001)	0.1037	0.0732
	Trained predictor (M0)	5	0.623 (0.008)	0.697 (0.006)	0.291 (0.017)	0.0274	0.0732
Organoids, primary pooled	Constant	5	undefined	undefined	undefined	8.72×10^{-18}	0.2407
	Training marginals	5	0.465 (0.001)	0.588 (0.001)	undefined	6.59×10^{-18}	0.2407
	Shuffled predictions	5	0.007 (0.015)	0.050 (0.065)	0.000 (0.013)	0.3878	0.2407

Supplementary Table 6: Cross-assay reproducibility of each response component, with cluster-bootstrap intervals. q_c is the correlation between the GDSC1 and GDSC2 measurements of the same drug-sample pair, estimated on the 32,724 matched pairs that meet the benchmark eligibility rules, over 89 compounds and 600 samples. Each bootstrap column gives the replicate mean and the 95% interval under one resampling scheme, 600 replicates each; the decomposition is rebuilt inside every replicate. $\sqrt{q_c}$ is the empirical reproducibility reference against which recovery is expressed; Supplementary Note 3 gives its assumptions.

Component	q_c	Two-way cluster	Compound cluster	Sample cluster	$\sqrt{q_c}$
Total response	0.650	0.636 [0.528, 0.750]	0.636 [0.529, 0.755]	0.650 [0.640, 0.660]	0.807
Additive component	0.754	0.728 [0.567, 0.866]	0.728 [0.562, 0.868]	0.754 [0.743, 0.764]	0.869
Interaction component	0.498	0.498 [0.444, 0.554]	0.499 [0.451, 0.547]	0.498 [0.481, 0.515]	0.706

Supplementary Table 7: Resource-composition sensitivity of the M2–M0 interaction effect. Differences are paired within seed. Positive gives the number of seeds whose interaction-correlation difference is positive. Each support contributes at least 1,000 test pairs and is a rescored subset of the same trained predictions.

Regime	Support	Mean pairs	Seeds	Mean Δr_{int}	s.d.	Positive	Mean ΔR_{int}^2
Held-out cell lines	NCI60	369,208	5	-0.0220	0.0368	1/5	-0.0139
	All except NCI60	155,147	5	0.0078	0.0227	4/5	0.0020
	PRISM	80,984	5	0.0240	0.0243	5/5	0.0223
	GDSC1	19,096	5	-0.0004	0.0070	2/5	-0.0039
	GDSC2	6,971	5	-0.0083	0.0406	3/5	-0.0229
	CTRP2	42,081	5	-0.0302	0.0514	1/5	-0.0466
	CTRP1	5,766	5	0.0090	0.0387	3/5	-0.0069
	CACLE	1,700	5	-0.0024	0.0233	2/5	-0.0047
Held-out compounds	NCI60	323,805	5	0.0237	0.0170	5/5	0.0216
	All except NCI60	145,156	5	0.0081	0.0039	5/5	0.0110
	PRISM	72,812	5	0.0181	0.0041	5/5	0.0264
	GDSC1	16,767	5	0.0040	0.0228	2/5	0.0102
	GDSC2	7,093	5	-0.0127	0.0310	1/5	0.0250
	CTRP2	40,817	5	0.0247	0.0251	4/5	0.0187
	CTRP1	6,111	5	0.0105	0.0200	3/5	0.0110
	CACLE	2,539	4	0.0137	0.0624	3/4	0.0121

Supplementary Table 8: Representation benchmark under held-out cell lines. Mean and standard deviation across seeds for every configuration, under a fixed nonlinear dual-tower decoder. *Full panel* records whether the representation covers the entire benchmark dataset; the two that do not are scored on the subset they cover. A_{int} is the emitted interaction amplitude. The model arms M0 and M2 are listed first and carry five seeds where the representations carry three. Expression is the untransformed gene-level expression input.

Class	Representation	Full panel	Seeds	Total	Additive	Interaction	A_{int}
Model arms	M0 (ECFP4 $\times$ Expression)	yes	5	0.858 (0.009)	0.963 (0.007)	0.315 (0.040)	0.346 (0.042)
	M2 (ECFP4 $\times$ Expression)	yes	5	0.858 (0.006)	0.963 (0.004)	0.307 (0.042)	0.376 (0.028)
Molecular representations	ECFP4	yes	3	0.859 (0.007)	0.964 (0.003)	0.308 (0.053)	0.356 (0.010)
	GeminiMol	no	3	0.828 (0.010)	0.933 (0.007)	0.283 (0.047)	0.323 (0.030)
	Mole_BERT	yes	3	0.827 (0.008)	0.929 (0.004)	0.274 (0.039)	0.338 (0.053)
	ChemBERTa2	yes	3	0.821 (0.011)	0.925 (0.007)	0.255 (0.057)	0.328 (0.014)
	MolT5	yes	3	0.829 (0.012)	0.934 (0.007)	0.249 (0.078)	0.324 (0.025)
	MolCLR	yes	3	0.742 (0.013)	0.834 (0.008)	0.238 (0.073)	0.231 (0.017)
	KPGT	yes	3	0.769 (0.016)	0.865 (0.019)	0.236 (0.069)	0.286 (0.037)
	UniMolV2	no	3	0.757 (0.028)	0.853 (0.029)	0.225 (0.035)	0.276 (0.039)
	UniMol	yes	3	0.768 (0.020)	0.868 (0.015)	0.208 (0.024)	0.292 (0.049)
	InfoAlign	yes	3	0.757 (0.022)	0.862 (0.015)	0.183 (0.106)	0.385 (0.031)
	Ouroboros	yes	3	0.678 (0.018)	0.766 (0.016)	0.170 (0.030)	0.222 (0.016)
	Chemprop	yes	3	0.513 (0.021)	0.587 (0.031)	0.118 (0.002)	0.284 (0.047)
Transcriptomic representations	SCimilarity	yes	3	0.865 (0.004)	0.969 (0.005)	0.318 (0.054)	0.350 (0.035)
	Expression	yes	3	0.859 (0.007)	0.964 (0.003)	0.308 (0.053)	0.356 (0.010)
	Tahoe_x1	yes	3	0.857 (0.007)	0.963 (0.004)	0.293 (0.050)	0.368 (0.029)
	scGPT	yes	3	0.860 (0.006)	0.967 (0.003)	0.290 (0.019)	0.374 (0.022)
	STATE	yes	3	0.849 (0.014)	0.959 (0.007)	0.276 (0.073)	0.433 (0.011)
	scFoundation	yes	3	0.858 (0.003)	0.967 (0.002)	0.272 (0.064)	0.376 (0.074)
	CellPLM	yes	3	0.857 (0.013)	0.964 (0.008)	0.271 (0.053)	0.306 (0.042)
	Cell2Sentence	yes	3	0.852 (0.015)	0.963 (0.012)	0.257 (0.034)	0.420 (0.035)
	BulkFormer	yes	3	0.849 (0.017)	0.960 (0.009)	0.252 (0.030)	0.356 (0.085)
	Geneformer	yes	3	0.853 (0.012)	0.965 (0.011)	0.243 (0.056)	0.392 (0.024)
	tGPT	yes	3	0.846 (0.015)	0.961 (0.009)	0.209 (0.020)	0.391 (0.017)
	UCE	yes	3	0.813 (0.006)	0.938 (0.015)	0.013 (0.049)	0.288 (0.069)

Supplementary Table 9: Representation benchmark under held-out compounds. Columns as in Supplementary Table 8.

Class	Representation	Full panel	Seeds	Total	Additive	Interaction	A_{int}
Model arms	M2 (ECFP4 $\times$ Expression)	yes	5	0.609 (0.013)	0.678 (0.014)	0.308 (0.013)	0.359 (0.012)
	M0 (ECFP4 $\times$ Expression)	yes	5	0.623 (0.008)	0.697 (0.006)	0.291 (0.017)	0.374 (0.007)
Molecular representations	ECFP4	yes	3	0.625 (0.010)	0.699 (0.008)	0.301 (0.014)	0.371 (0.002)
	GeminiMol	no	3	0.585 (0.016)	0.663 (0.016)	0.232 (0.012)	0.313 (0.033)
	Mole_BERT	yes	3	0.575 (0.017)	0.649 (0.016)	0.228 (0.011)	0.327 (0.025)
	MolCLR	yes	3	0.573 (0.009)	0.648 (0.011)	0.224 (0.017)	0.288 (0.007)
	MolT5	yes	3	0.531 (0.005)	0.599 (0.004)	0.214 (0.008)	0.296 (0.017)
	UniMolV2	no	3	0.565 (0.012)	0.640 (0.013)	0.207 (0.005)	0.280 (0.009)
	UniMol	yes	3	0.520 (0.019)	0.590 (0.020)	0.189 (0.003)	0.302 (0.013)
	Ouroboros	yes	3	0.537 (0.007)	0.611 (0.005)	0.186 (0.003)	0.302 (0.026)
	ChemBERTa2	yes	3	0.512 (0.003)	0.581 (0.006)	0.183 (0.005)	0.316 (0.038)
	KPGT	yes	3	0.360 (0.021)	0.411 (0.024)	0.097 (0.011)	0.186 (0.043)
	Chemprop	yes	3	0.173 (0.014)	0.201 (0.015)	0.006 (0.011)	0.058 (0.011)
	InfoAlign	yes	3	0.160 (0.007)	0.187 (0.007)	-0.001 (0.005)	0.067 (0.004)
Transcriptomic representations	STATE	yes	3	0.628 (0.009)	0.702 (0.007)	0.303 (0.019)	0.400 (0.019)
	scGPT	yes	3	0.625 (0.012)	0.698 (0.012)	0.303 (0.013)	0.378 (0.009)
	Expression	yes	3	0.625 (0.010)	0.699 (0.008)	0.301 (0.014)	0.371 (0.002)
	Tahoe_x1	yes	3	0.623 (0.008)	0.697 (0.008)	0.300 (0.003)	0.384 (0.033)
	scFoundation	yes	3	0.620 (0.007)	0.693 (0.007)	0.296 (0.009)	0.376 (0.022)
	BulkFormer	yes	3	0.620 (0.017)	0.694 (0.016)	0.293 (0.012)	0.384 (0.015)
	SCimilarity	yes	3	0.613 (0.012)	0.686 (0.013)	0.290 (0.006)	0.353 (0.021)
	tGPT	yes	3	0.615 (0.016)	0.689 (0.017)	0.287 (0.016)	0.376 (0.023)
	CellPLM	yes	3	0.614 (0.016)	0.687 (0.015)	0.285 (0.017)	0.364 (0.024)
	Cell2Sentence	yes	3	0.618 (0.009)	0.693 (0.007)	0.280 (0.011)	0.342 (0.018)
	UCE	yes	3	0.618 (0.006)	0.693 (0.006)	0.276 (0.011)	0.355 (0.030)
	Geneformer	yes	3	0.610 (0.011)	0.685 (0.011)	0.268 (0.013)	0.339 (0.020)

Supplementary Table 10: Descriptive score separation in the representation benchmark. Spread is the range of the mean score across representations on one axis, median seed s.d. is the median across representations of the seed-to-seed standard deviation, and Ratio is their quotient, which describes score separation relative to seed variation. Without UCE rows use the other eleven transcriptomic representations.

Regime	Axis	Component	Spread	Median seed s.d.	Ratio
Held-out cell lines	Molecular	Total response	0.3464	0.0147	23.6
		Additive component	0.3771	0.0114	33.0
		Interaction component	0.1897	0.0550	3.5
	Transcriptomic	Total response	0.0520	0.0097	5.4
		Additive component	0.0310	0.0079	3.9
		Interaction component	0.3055	0.0514	5.9
	Transcriptomic, without UCE	Total response	0.0192	0.0123	1.6
		Additive component	0.0097	0.0074	1.3
		Interaction component	0.1092	0.0530	2.1
Held-out compounds	Molecular	Total response	0.4650	0.0097	47.9
		Additive component	0.5119	0.0095	54.0
		Interaction component	0.3020	0.0095	31.9
	Transcriptomic	Total response	0.0186	0.0106	1.8
		Additive component	0.0175	0.0098	1.8
		Interaction component	0.0352	0.0126	2.8
	Transcriptomic, without UCE	Total response	0.0186	0.0111	1.7
		Additive component	0.0175	0.0115	1.5
		Interaction component	0.0352	0.0129	2.7

Supplementary Table 11: Component-wise recovery under three decoders. Mean and standard deviation across the runs of each configuration. The additive ridge decoder is additive by construction, so its prediction lies in the additive subspace and it emits an interaction of standard deviation 10^{-9} to 10^{-8} , against an observed interaction standard deviation of order 7×10^{-2} ; its interaction-component correlations are correlations of floating-point noise (Supplementary Note 2). Three of the sixty low-rank bilinear runs, all at the largest weight decay, collapse to the same condition. $\text{sd}(\hat{M}\hat{y})$ is the emitted interaction amplitude in absolute units.

Decoder	Molecular	Transcriptomic	Runs	Parameters	Total	Additive	Interaction	$\text{sd}(\hat{M}\hat{y})$
Additive ridge	ECFP4	BulkFormer	4	2,689	0.408 (0.103)	0.461 (0.117)	0.009 (0.007)	0.0000
	ECFP4	CellPLM	4	2,561	0.483 (0.111)	0.546 (0.125)	0.005 (0.004)	0.0000
	ECFP4	Expression	4	18,010	0.254 (0.183)	0.287 (0.206)	0.004 (0.006)	0.0000
	ECFP4	scGPT	4	2,561	0.470 (0.094)	0.531 (0.106)	-0.007 (0.010)	0.0000
	KPGT	Expression	4	18,266	0.176 (0.067)	0.199 (0.076)	-0.001 (0.002)	0.0000
	MolCLR	Expression	4	16,474	0.099 (0.094)	0.112 (0.106)	-0.000 (0.001)	0.0000
	UniMol	Expression	4	16,474	0.151 (0.124)	0.171 (0.140)	0.000 (0.001)	0.0000
Low-rank bilinear	ECFP4	BulkFormer	12	88,705	0.351 (0.114)	0.419 (0.118)	0.051 (0.040)	0.0559
	ECFP4	CellPLM	12	84,481	0.493 (0.106)	0.569 (0.126)	0.080 (0.053)	0.0185
	ECFP4	Expression	12	594,298	0.266 (0.147)	0.304 (0.170)	0.047 (0.038)	0.0501
	ECFP4	scGPT	12	84,481	0.391 (0.104)	0.466 (0.108)	0.058 (0.050)	0.0431
	KPGT	Expression	4	602,746	0.181 (0.087)	0.207 (0.100)	0.024 (0.021)	0.0581
	MolCLR	Expression	4	543,610	0.074 (0.075)	0.084 (0.085)	0.009 (0.009)	0.0632
	UniMol	Expression	4	543,610	0.125 (0.092)	0.141 (0.105)	0.018 (0.027)	0.0370
Nonlinear dual tower	ECFP4	BulkFormer	3	3,812,353	0.845 (0.011)	0.951 (0.009)	0.292 (0.010)	0.0282
	ECFP4	CellPLM	3	3,681,281	0.856 (0.020)	0.962 (0.015)	0.275 (0.049)	0.0224
	ECFP4	Expression	3	19,501,057	0.856 (0.011)	0.957 (0.007)	0.332 (0.064)	0.0262
	ECFP4	scGPT	3	3,681,281	0.857 (0.008)	0.962 (0.007)	0.300 (0.028)	0.0257
	KPGT	Expression	1	19,632,129	0.836	0.933	0.278	0.0202
	MolCLR	Expression	1	18,714,625	0.817	0.911	0.288	0.0209
	UniMol	Expression	1	18,714,625	0.837	0.933	0.294	0.0219

Supplementary Table 12: Agreement between representation rankings under different decoders. ρ_s is the Spearman correlation between the two rankings named in Comparison, over the representations that both decoders were run on. Top 1 and Top 2 record whether the leading representation and the leading pair are the same under both. Molecular rows hold gene-level expression fixed and transcriptomic rows hold ECFP4 fixed. Seed 3407 rows compare the orderings of four representations under two decoders, as in Fig. 3f; three-seed rows compare a decoder with itself across three seeds and give its mean ordering. Comparisons on the interaction component are omitted where one of the two decoders emits no interaction.

Scope	Axis	Component	Comparison	ρ_s	Top 1	Top 2	Mean ordering
Seed 3407	Molecular	Total	ridge vs bilinear	0.80	yes	no	
Seed 3407	Molecular	Total	ridge vs tower	1.00	yes	yes	
Seed 3407	Molecular	Total	bilinear vs tower	0.80	yes	no	
Seed 3407	Molecular	Additive	ridge vs bilinear	0.80	yes	no	
Seed 3407	Molecular	Additive	ridge vs tower	1.00	yes	yes	
Seed 3407	Molecular	Additive	bilinear vs tower	0.80	yes	no	
Seed 3407	Molecular	Interaction	bilinear vs tower	0.20	yes	no	
Seed 3407	Transcriptomic	Total	ridge vs bilinear	0.80	no	yes	
Seed 3407	Transcriptomic	Total	ridge vs tower	0.40	yes	no	
Seed 3407	Transcriptomic	Total	bilinear vs tower	0.00	no	no	
Seed 3407	Transcriptomic	Additive	ridge vs bilinear	1.00	yes	yes	
Seed 3407	Transcriptomic	Additive	ridge vs tower	0.80	yes	yes	
Seed 3407	Transcriptomic	Additive	bilinear vs tower	0.80	yes	yes	
Seed 3407	Transcriptomic	Interaction	bilinear vs tower	-0.80	no	no	
Three seeds	Transcriptomic	Total	bilinear, seed stability	0.87	no	—	scGPT > CellPLM > BulkFormer > Expression
Three seeds	Transcriptomic	Additive	bilinear, seed stability	1.00	yes	—	CellPLM > scGPT > BulkFormer > Expression
Three seeds	Transcriptomic	Interaction	bilinear, seed stability	1.00	yes	—	scGPT > CellPLM > BulkFormer > Expression
Three seeds	Transcriptomic	Total	tower, seed stability	-0.20	no	—	scGPT > Expression > CellPLM > BulkFormer
Three seeds	Transcriptomic	Additive	tower, seed stability	0.40	no	—	CellPLM > scGPT > Expression > BulkFormer
Three seeds	Transcriptomic	Interaction	tower, seed stability	-0.33	no	—	Expression > scGPT > BulkFormer > CellPLM

Supplementary Table 13: Component-wise error geometry, per run. d_H is the dimension of the additive subspace and d_H/N its share of the degrees of freedom, in per cent. f_H is the share of total squared prediction error that falls in the additive subspace. D_H and D_M are the rank-normalized error densities of the two subspaces and $C_{\text{error}} = D_H/D_M$ their ratio. E_{add} and E_{int} are the two error terms of the exact attribution $\text{MSE} = E_{\text{add}} + E_{\text{int}}$. The three error quantities are given in units of 10^{-3} .

Regime	Arm	Seed	N	d_H	d_H/N	f_H	D_H	D_M	C_{error}	E_{add}	E_{int}
Held-out cell lines	M0	3407	557,484	54,323	9.74	0.257	15.82	4.93	3.21	1.54	4.45
	M0	3408	474,279	54,323	11.45	0.222	11.44	5.19	2.20	1.31	4.59
	M0	3409	486,991	54,319	11.15	0.207	11.29	5.42	2.08	1.26	4.82
	M0	3410	570,863	54,322	9.52	0.303	19.99	4.83	4.14	1.90	4.37
	M0	3411	530,351	54,322	10.24	0.182	9.92	5.09	1.95	1.02	4.57
	M2	3407	557,484	54,323	9.74	0.255	15.75	4.96	3.18	1.53	4.47
	M2	3408	474,279	54,323	11.45	0.239	12.48	5.13	2.43	1.43	4.54
	M2	3409	486,991	54,319	11.15	0.206	11.25	5.43	2.07	1.25	4.83
	M2	3410	570,863	54,322	9.52	0.247	15.17	4.86	3.12	1.44	4.40
	M2	3411	530,351	54,322	10.24	0.195	11.19	5.28	2.12	1.15	4.74
Held-out compounds	M0	3407	468,871	9,112	1.94	0.632	431.36	4.97	86.81	8.38	4.87
	M0	3408	466,230	9,110	1.95	0.634	448.38	5.15	87.04	8.76	5.05
	M0	3409	459,058	9,110	1.98	0.642	449.94	5.08	88.52	8.93	4.98
	M0	3410	475,001	9,112	1.92	0.643	472.71	5.12	92.28	9.07	5.02
	M0	3411	474,086	9,110	1.92	0.641	444.60	4.88	91.19	8.54	4.78
	M2	3407	468,871	9,112	1.94	0.651	459.78	4.88	94.31	8.94	4.78
	M2	3408	466,230	9,110	1.95	0.655	484.06	5.09	95.14	9.46	4.99
	M2	3409	459,058	9,110	1.98	0.652	467.14	5.05	92.45	9.27	4.95
	M2	3410	475,001	9,112	1.92	0.640	455.80	5.01	91.04	8.74	4.91
	M2	3411	474,086	9,110	1.92	0.658	468.39	4.78	97.99	9.00	4.69

Supplementary Table 14: Recovery relative to the empirical reproducibility reference, per run. $\eta_c = r_c / \sqrt{q_c}$ on the matched support that each regime holds out, whose size is given in the fourth column. The held-out compound rows rest on 7 to 19 held-out compounds (Supplementary Note 3).

Regime	Arm	Seed	Matched pairs	η_{total}	η_{add}	η_{int}
Held-out cell lines	M0	3407	2,968	0.861	1.032	0.358
	M0	3408	3,015	0.829	1.009	0.365
	M0	3409	2,904	0.831	1.011	0.361
	M0	3410	3,051	0.815	0.973	0.391
	M0	3411	2,876	0.858	0.997	0.375
	M2	3407	2,968	0.833	1.002	0.334
	M2	3408	3,015	0.822	1.003	0.375
	M2	3409	2,904	0.824	1.008	0.350
	M2	3410	3,051	0.820	0.980	0.412
	M2	3411	2,876	0.855	0.986	0.406
Held-out compounds	M0	3407	2,721	0.297	0.322	0.120
	M0	3408	3,470	0.365	0.443	0.261
	M0	3409	1,970	0.552	0.564	0.538
	M0	3410	3,873	0.590	0.646	0.221
	M0	3411	2,998	0.301	0.338	0.032
	M2	3407	2,721	0.344	0.366	0.162
	M2	3408	3,470	0.380	0.476	0.239
	M2	3409	1,970	0.263	0.177	0.675
	M2	3410	3,873	0.574	0.630	0.234
	M2	3411	2,998	0.140	0.163	0.005

Supplementary Table 15: Input-compatibility check, per organoid cohort. Level and s.d. summarize the distribution of the transcriptomic input over each cohort, Gap vs CCLE is the absolute difference in level from the training reference in the first row, and Zeros is the fraction of entries equal to zero. Samples counts the expression profiles of each cohort, including organoids with no evaluated response pair; the 173 organoids of the zero-shot evaluation (Fig. 1b) are those with at least one, and Rows counts their drug–organoid pairs. *Primary* marks the primary set (Methods), and *Compatible* marks cohorts that meet all three conditions.

Cohort	Samples	Genes	Level	s.d.	Gap vs CCLE	Zeros	Min	Primary	Compatible	Rows
CCLE (training reference)	—	—	2.702	0.221	0.000	0.153	0.00	—	—	—
HKUPDO	52	8,866	0.008	0.075	2.694	0.951	-9.97	no	no	324
LICOB	65	13,224	0.399	0.015	2.303	0.000	-1.80	no	no	4,800
UMPDO1	90	15,413	3.578	0.128	0.876	0.114	0.00	yes	yes	2,744
UMPDO2	34	15,218	3.397	0.159	0.694	0.158	0.00	yes	yes	1,470
UMPDO3	14	14,651	3.707	0.198	1.005	0.094	0.00	yes	no	672

Supplementary Table 16: Matched three-arm comparison. M0, M1 and M2 were trained on the same dataset, data partitions and five seeds and selected by minimum validation total-response error. M1 and M2 have identical architecture and parameter count; only M2 receives component-wise supervision. Entries are mean (s.d.) over seeds.

Regime	Arm	Seeds	Mean pairs	Total	Additive	Interaction	A_{int}	Interaction R^2	Parameters
Held-out cell lines	M0	5	523,994 0.858 (0.009)	0.963 (0.007)	0.315 (0.040)	0.346 (0.042)	0.097 (0.029)	19,501,057	
	M1	5	523,994 0.855 (0.010)	0.960 (0.007)	0.306 (0.044)	0.369 (0.030)	0.090 (0.031)	20,029,444	
	M2	5	523,994 0.858 (0.006)	0.963 (0.004)	0.307 (0.042)	0.376 (0.028)	0.090 (0.032)	20,029,444	
Held-out compounds	M0	5	468,649 0.623 (0.008)	0.697 (0.006)	0.291 (0.017)	0.374 (0.007)	0.078 (0.014)	19,501,057	
	M1	5	468,649 0.618 (0.013)	0.692 (0.012)	0.279 (0.018)	0.354 (0.023)	0.072 (0.012)	20,029,444	
	M2	5	468,649 0.609 (0.013)	0.678 (0.014)	0.308 (0.013)	0.359 (0.012)	0.092 (0.009)	20,029,444	

Supplementary Table 17: Model choices and their independent GDSC2 endpoints. Each criterion selects M0 or M2 on GDSC1 separately by split and seed; M2 of 5 is the number of M2 choices. Δ MAE is the paired independent GDSC2 mean absolute error relative to global-correlation selection, so positive is worse. Sign, PCC and Spearman are secondary endpoints.

Regime	Selection criterion	M2 of 5	GDSC2 MAE	Δ MAE	Sign	PCC	Spearman
Held-out cell lines	Global	1	0.13716	0.00000	0.5767	0.2815	0.2490
	Fixed drug	4	0.14079	0.00363	0.5778	0.2795	0.2489
	Fixed cell	1	0.13716	0.00000	0.5767	0.2815	0.2490
	DrEval normalized	1	0.13716	0.00000	0.5767	0.2815	0.2490
	DrugDis interaction correlation	3	0.13961	0.00245	0.5792	0.2879	0.2558
	DrugDis interaction R^2	1	0.13716	0.00000	0.5767	0.2815	0.2490
Held-out compounds	Global	2	0.12980	0.00000	0.5636	0.2979	0.2269
	Fixed drug	2	0.13056	0.00076	0.5651	0.2900	0.2321
	Fixed cell	2	0.12980	0.00000	0.5636	0.2979	0.2269
	DrEval normalized	2	0.12980	0.00000	0.5636	0.2979	0.2269
	DrugDis interaction correlation	4	0.13023	0.00043	0.5655	0.3094	0.2334
	DrugDis interaction R^2	3	0.13009	0.00029	0.5662	0.3018	0.2330

Supplementary Table 18: Amplitude-sensitive interaction variance explained. Interaction is the component correlation, A_{int} is the emitted amplitude relative to the observed interaction, and $R_{\text{int}}^2 = 2A_{\text{int}}r_{\text{int}} - A_{\text{int}}^2$. Entries are mean (s.d.) over five runs selected by minimum validation mean squared error. Unlike a squared correlation, this quantity accounts for amplitude and can be negative.

Regime	Arm	Seeds	Interaction	A_{int}	R_{int}^2
Held-out cell lines	M0	5	0.315 (0.040)	0.346 (0.042)	0.097 (0.029)
	M2	5	0.307 (0.042)	0.376 (0.028)	0.090 (0.032)
Held-out compounds	M0	5	0.291 (0.017)	0.374 (0.007)	0.078 (0.014)
	M2	5	0.308 (0.013)	0.359 (0.012)	0.092 (0.009)
Organoids, primary pooled	M0	5	0.067 (0.010)	0.179 (0.049)	-0.010 (0.012)
	M2	5	0.087 (0.009)	0.209 (0.021)	-0.008 (0.007)

Supplementary Table 19: Independent assay-generation selectivity evaluation by model arm. GDSC1 is the model-selection assay generation and GDSC2 the independent test generation. Each split and seed uses 50,000 support-sampled double-difference rectangles. Values are means over five seeds.

Regime	Arm	Seeds	Mean pairs	GDSC1 MAE	GDSC1 sign	GDSC1 PCC	GDSC2 MAE	GDSC2 sign	GDSC2 PCC
Held-out cell lines	M0	5	5,138	0.1946	0.5696	0.2315	0.1371	0.5782	0.2817
	M2	5	5,138	0.1979	0.5694	0.2252	0.1429	0.5747	0.2629
Held-out compounds	M0	5	5,027	0.1804	0.5525	0.2133	0.1303	0.5624	0.2837
	M2	5	5,027	0.1794	0.5572	0.2361	0.1303	0.5651	0.3055

Supplementary Table 20: The component-supervised predictor against the baseline, per seed. M0 is the total-response-supervised baseline. M2 adds two effect heads, component-wise supervision and 2.7% trainable parameters. Arms are paired within seed, on the same split. MSE, E_{add} and E_{int} are given in units of 10^{-3} and satisfy $\text{MSE} = E_{\text{add}} + E_{\text{int}}$ exactly. Epoch is the selected checkpoint, the epoch of lowest validation total prediction error.

Regime	Arm	Seed	N	Total	Additive	Interaction	A_{int}	MSE	E_{add}	E_{int}	Epoch	Parameters
Held-out cell lines	M0	3407	557,484	0.8671	0.9661	0.3160	0.346	5.99	1.54	4.45	16	19,501,057
	M0	3408	474,279	0.8566	0.9608	0.3561	0.356	5.90	1.31	4.59	24	19,501,057
	M0	3409	486,991	0.8535	0.9642	0.2507	0.365	6.07	1.26	4.82	30	19,501,057
	M0	3410	570,863	0.8466	0.9514	0.3173	0.277	6.27	1.90	4.37	12	19,501,057
	M0	3411	530,351	0.8667	0.9714	0.3371	0.388	5.59	1.02	4.57	30	19,501,057
	M2	3407	557,484	0.8653	0.9654	0.3101	0.364	6.01	1.53	4.47	19	20,029,444
	M2	3408	474,279	0.8604	0.9640	0.3724	0.416	5.97	1.43	4.54	24	20,029,444
	M2	3409	486,991	0.8547	0.9648	0.2570	0.394	6.08	1.25	4.83	27	20,029,444
	M2	3410	570,863	0.8501	0.9556	0.3087	0.357	5.84	1.44	4.40	20	20,029,444
	M2	3411	530,351	0.8588	0.9668	0.2886	0.351	5.88	1.15	4.74	19	20,029,444
Held-out compounds	M0	3407	468,871	0.6157	0.6925	0.2869	0.373	13.26	8.38	4.87	22	19,501,057

Supplementary Table 20: Continued.

Regime	Arm	Seed	<i>N</i>	Total	Additive	Interaction	<i>A</i> _{int}	MSE	<i>E</i> _{add}	<i>E</i> _{int}	Epoch	Parameters
	M0	3408	466,230	0.6358	0.7083	0.3154	0.369	13.81	8.76	5.05	25	19,501,057
	M0	3409	459,058	0.6244	0.6965	0.3012	0.370	13.91	8.93	4.98	22	19,501,057
	M0	3410	475,001	0.6201	0.6953	0.2750	0.372	14.09	9.07	5.02	14	19,501,057
	M0	3411	474,086	0.6195	0.6940	0.2762	0.386	13.33	8.54	4.78	26	19,501,057
	M2	3407	468,871	0.5900	0.6584	0.3086	0.362	13.72	8.94	4.78	22	20,029,444
	M2	3408	466,230	0.6144	0.6806	0.3290	0.351	14.45	9.46	4.99	21	20,029,444
	M2	3409	459,058	0.6126	0.6817	0.3051	0.346	14.22	9.27	4.95	16	20,029,444
	M2	3410	475,001	0.6233	0.6960	0.3044	0.376	13.65	8.74	4.91	28	20,029,444
	M2	3411	474,086	0.6034	0.6723	0.2947	0.361	13.69	9.00	4.69	30	20,029,444

Supplementary Table 21: Zero-shot organoid evaluation, pooled and per cohort. Mean and standard deviation across five seeds. The first block is the primary analysis over UMPDO1–3; the second pools all five cohorts; the remaining rows give each cohort separately, with its Primary and Compatible flags.

Cohort	Primary	Compatible	Arm	Seeds	<i>N</i>	Total	Additive	Interaction	<i>A</i> _{int}
Primary pooled			M0	5	4,886	0.460 (0.007)	0.607 (0.011)	0.067 (0.010)	0.179 (0.049)
			M2	5	4,886	0.466 (0.036)	0.614 (0.046)	0.087 (0.009)	0.209 (0.021)
All cohorts, pooled			M0	5	10,010	0.271 (0.097)	0.356 (0.128)	0.040 (0.030)	0.206 (0.041)
			M2	5	10,010	0.382 (0.048)	0.500 (0.058)	0.078 (0.026)	0.235 (0.015)
HKUPDO	no	no	M0	5	324	0.442 (0.088)	0.493 (0.106)	-0.005 (0.092)	0.563 (0.165)
	no	no	M2	5	324	0.358 (0.096)	0.402 (0.106)	-0.015 (0.096)	0.736 (0.051)
LICOB	no	no	M0	5	4,800	0.268 (0.045)	0.377 (0.063)	0.066 (0.019)	0.118 (0.096)
	no	no	M2	5	4,800	0.283 (0.035)	0.402 (0.036)	0.054 (0.066)	0.224 (0.075)
UMPDO1	yes	yes	M0	5	2,744	0.561 (0.014)	0.662 (0.015)	0.054 (0.017)	0.221 (0.049)
	yes	yes	M2	5	2,744	0.547 (0.034)	0.648 (0.036)	0.077 (0.013)	0.279 (0.023)
UMPDO2	yes	yes	M0	5	1,470	0.358 (0.011)	0.552 (0.017)	0.058 (0.019)	0.092 (0.017)
	yes	yes	M2	5	1,470	0.364 (0.023)	0.560 (0.037)	0.069 (0.014)	0.097 (0.024)
UMPDO3	yes	no	M0	5	672	0.383 (0.024)	0.461 (0.038)	0.140 (0.052)	0.147 (0.036)
	yes	no	M2	5	672	0.391 (0.048)	0.466 (0.049)	0.167 (0.107)	0.187 (0.009)

Supplementary Table 22: Zero-shot organoid evaluation of the matched control M1. Held-out-cell-line checkpoints were applied to the organoids. The primary pooled set is UMPDO1, UMPDO2 and UMPDO3, and M0 and M2 are shown for the three-arm comparison. Entries are mean (s.d.) over five seeds.

Cohort	Arm	Seeds	<i>N</i>	Additive	Interaction	<i>A</i> _{int}	Interaction <i>R</i> ²
Primary pooled	M0	5	4,886	0.607 (0.011)	0.067 (0.010)	0.179 (0.049)	-0.010 (0.012)
	M1	5	4,886	0.621 (0.024)	0.061 (0.013)	0.165 (0.059)	-0.009 (0.011)
	M2	5	4,886	0.614 (0.046)	0.087 (0.009)	0.209 (0.021)	-0.008 (0.007)
UMPDO1	M0	5	2,744	0.662 (0.015)	0.054 (0.017)	0.221 (0.049)	-0.028 (0.020)
	M1	5	2,744	0.653 (0.028)	0.048 (0.026)	0.213 (0.099)	-0.031 (0.032)
	M2	5	2,744	0.648 (0.036)	0.077 (0.013)	0.279 (0.023)	-0.035 (0.010)
UMPDO2	M0	5	1,470	0.552 (0.017)	0.058 (0.019)	0.092 (0.017)	0.002 (0.003)
	M1	5	1,470	0.548 (0.025)	0.052 (0.018)	0.074 (0.020)	0.002 (0.004)
	M2	5	1,470	0.560 (0.037)	0.069 (0.014)	0.097 (0.024)	0.004 (0.002)
UMPDO3	M0	5	672	0.461 (0.038)	0.140 (0.052)	0.147 (0.036)	0.019 (0.013)
	M1	5	672	0.483 (0.051)	0.115 (0.089)	0.136 (0.059)	0.015 (0.017)
	M2	5	672	0.466 (0.049)	0.167 (0.107)	0.187 (0.009)	0.028 (0.040)
All cohorts, pooled	M0	5	10,010	0.356 (0.128)	0.040 (0.030)	0.206 (0.041)	-0.027 (0.016)
	M1	5	10,010	0.354 (0.150)	0.032 (0.031)	0.186 (0.038)	-0.023 (0.009)
	M2	5	10,010	0.500 (0.058)	0.078 (0.026)	0.235 (0.015)	-0.019 (0.011)
LICOB	M0	5	4,800	0.377 (0.063)	0.066 (0.019)	0.118 (0.096)	-0.005 (0.021)
	M1	5	4,800	0.410 (0.026)	0.072 (0.048)	0.106 (0.059)	0.002 (0.013)
	M2	5	4,800	0.402 (0.036)	0.054 (0.066)	0.224 (0.075)	-0.032 (0.047)
HKUPDO	M0	5	324	0.493 (0.106)	-0.005 (0.092)	0.563 (0.165)	-0.326 (0.154)
	M1	5	324	0.468 (0.096)	0.013 (0.054)	0.690 (0.166)	-0.482 (0.264)
	M2	5	324	0.402 (0.106)	-0.015 (0.096)	0.736 (0.051)	-0.565 (0.146)

Supplementary Data

Supplementary Data 1: Data sources

Supplementary Data 1 lists each response resource and organoid cohort used in this study, with its use, the DROMA tables read, the rows used and its primary publication. All data were read from the processed DROMA collection ([10.5281/zenodo.18503188](https://doi.org/10.5281/zenodo.18503188)). Rows are response rows in the benchmark dataset for a resource and rows scored in the zero-shot cross-system evaluation for a cohort.

Resource or cohort	Use in this study	Rows	Primary publication
NCI60	Benchmark dataset	2,175,581	Holbeck et al., Mol. Cancer Ther. (2010)
PRISM	Benchmark dataset	504,493	Corsello et al., Nat. Cancer (2020)
CTRP2	Benchmark dataset	269,725	Seashore-Ludlow et al., Cancer Discov. (2015)
GDSC1	Benchmark dataset; cross-assay reproducibility; model-selection test (selection arm)	125,914	Garnett et al., Nature (2012)
GDSC2	Benchmark dataset; cross-assay reproducibility; model-selection test (test arm)	44,797	Iorio et al., Cell (2016)
CTRP1	Benchmark dataset	39,784	Basu et al., Cell (2013)
CCLE	Benchmark dataset; transcriptomic input matrix	10,730	Barretina et al., Nature (2012)
GRAY	Benchmark dataset	1,888	Daemen et al., Genome Biol. (2013)
gCSI	Benchmark dataset	1,646	Haverty et al., Nature (2016)
FIMM	Benchmark dataset	890	Pemovska et al., Cancer Discov. (2013)
UHNBreast	Benchmark dataset	54	Marcotte et al., Cell (2016)
UMPDO1	Zero-shot evaluation, primary set	2,744	—
UMPDO2	Zero-shot evaluation, primary set	1,470	—
UMPDO3	Zero-shot evaluation, primary set	672	—
LICOB	Zero-shot evaluation, input-incompatible	4,800	Ji et al., Sci. Transl. Med. (2023)
HKUPDO	Zero-shot evaluation, input-incompatible	324	—